Spatial Compilation of Stochastic Chemical Trajectories

Harjas Sandhu
The University of Queensland
harjas.sandhu@student.uq.edu.au

13 November 2026

Abstract

A well-mixed reaction specification does not by itself determine a local particle implementation. We construct such an implementation for finite labelled mass-action networks with a positive integer conservation law, a supplied reverse involution and prescribed rational detailed-balanced activities. The compiler produces elementary reactions, strictly positive particle radii, constant rational coefficients, a payload decoder and one known slowdown. From every feasible initial placement, including simultaneous contacts, the decoded timed and labelled path approximates the target path in total variation. Transport may be unknown, spatially inhomogeneous and time dependent; the estimate permits full-history predictable chemical, drift and correlated covariance faults within a computed tolerance. A dilute hard-ball Poincaré inequality, proved by buffering, rank separation and one-particle coordinate paths, supplies the required population-uniform polynomial mixing estimates. We account for duration, inverse radii and coefficients, precision, every attempted reaction and the actual wall and pair pushes. The bounds are polynomial in a stated numerical charge together with the binary input length. They allow severe dilution and slowdown and concern an ideal particle model, without asserting a molecular realization or a practical advantage.

Contents

1 From a reaction specification to moving particles . . . 2
2 Model and main theorem . . . 4
3 Elementary chemistry and exact spatial balance . . . 6
4 Dilute hard-ball geometry . . . 9
5 Unknown transport and conditional mixing . . . 13
6 Comparison of complete timed paths . . . 16
7 Compiler choices and all-run resources . . . 18
8 Scope, obstructions and further questions . . . 20
Acknowledgements and AI use . . . 20
A Reflected motion: estimates, construction and smoothing . . . 21
B Observable tubes and conditional telescoping . . . 25
C Finite-bit geometry and polynomial majorants . . . 27
D Exact import contracts and dependence . . . 28
References . . . 30

1 From a reaction specification to moving particles

A mass-action model specifies counts and reaction events while suppressing positions, encounter delays and rejected product placements. Can such a table be compiled into a local particle system that reproduces the full stochastic record, with one known change of clock and explicit resource bounds?

Here the target is finite, conservative and reversible; its reaction order, species and labels may vary. The vessel is a three-dimensional rational convex polytope and every particle has positive radius. The compiler uses bounds on unknown species-dependent transport and predictable perturbations, without measuring initial positions or solving a configuration-space equation.

Accuracy concerns the joint law of initial counts, continuous firing times, original labels and post-event counts, including null events. Total variation therefore controls every measurable question about this finite-horizon record, rather than just a single-time distribution. The clock change is deterministic; no internal waiting time is removed. The path laws are approximated, not equal.

Contribution and relation to the kinetic companion. The elementary well-mixed compiler in [6, Theorem 9.1] is a separate conservative, detailed-balanced construction. We use precisely that theorem, with a smaller error budget. Its general open-network theorem has a different input class and does not supply detailed balance. Our additional operation is a finite-radius continuum realization with quantitative control of transport, spatial rejection and the complete marked path. The central geometric estimate is

$$\operatorname{Var}_{\mu_{\mathcal{K}}} f \le 2^{20}(1+N)\int_{\mathcal{K}} |\nabla f|^2 \, d\mu_{\mathcal{K}},$$

for every labelled configuration space beneath the conserved cap, when the computed radius is sufficiently small. Its proof avoids exponential congestion in the number of particles. Local reflected estimates then give all-start mixing and conditional comparison under history-dependent faults. Finally, a category coupling preserves the actual hidden history while matching the exact proposal times.

Predecessors. Programmable DNA controllers have experimental realizations with measured performance [2]. We instead specify an ideal stochastic model. Elimination of fast intermediates is established: Hoessly–Wiuf [5, Assumption 2 and Theorem 4.1] give a finite-closed-set reduction controlling compact-time state probabilities. That statement alone does not provide the full marked-path and resource contract used here. Poisson product-form laws and their restrictions to chemical classes are classical under appropriate balance hypotheses [1, Theorem 4.1 and Section 4.2].

Spatial detailed balance depends on placement as well as reaction rates. Zhang–Isaacson [7, Section III and Theorem III.1] give the relevant forward/reverse flux criterion for reversible particle reactions, including boundary effects. We apply it with anchored placement and steric rejection. For reflected hard particles and contact-clock decomposition, see Fradon [4, Theorem 2.2 and Proposition 3.4]. Its normal-reflection theorems are not invoked as a theorem about arbitrary variable conormal reflection. We supply that passage for the present geometry. The smooth-boundary derivative formula used for local smoothing is Cheng–Zhang [3, Theorem 2.5], with its curvature and local-time hypotheses checked in appendix D.

The contribution combines these components with the dilute hard-ball comparison, conditional robustness and acyclic parameter choices needed for a variable-network compiler. Historical firstness is not established here.

1.1 Conceptual overview

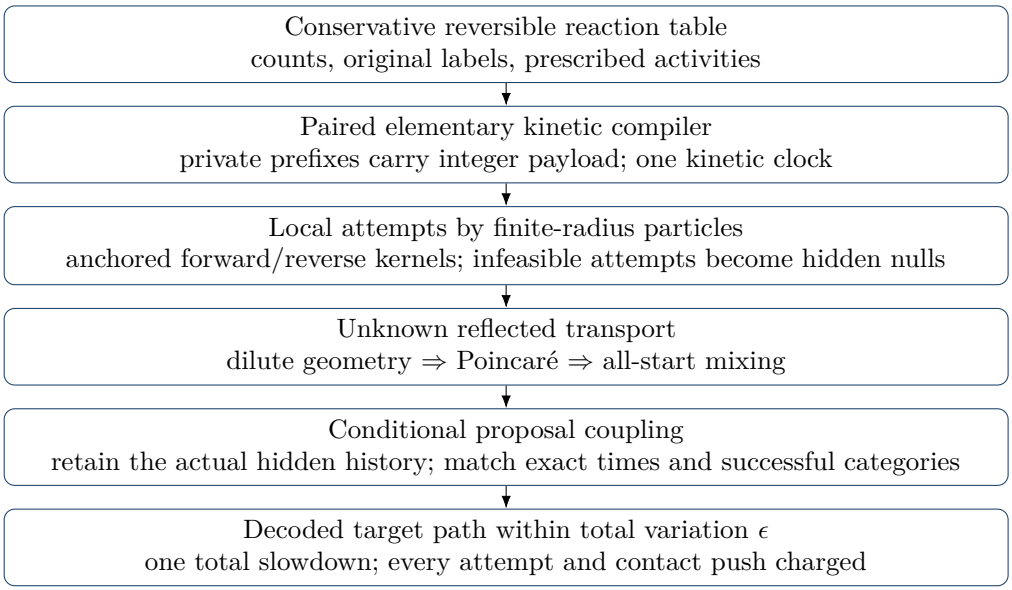


Figure 1: The compilation argument. Buffers, rank maps, metric projections and random coordinate paths occur only in proofs. They are not forces or relocation operations in the emitted system.

Three issues require care. First, exact balance is a local flux identity, whereas accurate well-mixed kinetics is an averaging statement. The former holds exactly for successful nominal jumps; the latter pays the small excluded-volume bias. Second, small excluded volume does not by itself prove mixing from a touching configuration. The hard-ball Poincaré proof and local reflection estimates address that distinct problem. Third, a perturbation policy may read every hidden event. Replacing the physical history by fresh equilibrium positions would change the policy. Our comparison changes only an auxiliary reference category, leaving the actual physical history intact.

The radius is chosen before the extra spatial slowdown. The radius pays stationary bias per proposal; the slowdown pays relaxation between proposals. Thus the proof never asks a radius chosen at the start to be small relative to a later, radius-dependent horizon.

Symbol	Meaning
x, X_t	target count and reference-time target process
Z_s, Q_s	fine counts and positions at physical time s
P, d_s	payload matrix and unweighted payload size $d_s = \lvert P_s\rvert_1$
N, H	conserved mass cap and total physical slowdown
ρ, r	particle-size scale and encounter/placement range
A_0, A	prescribed nominal block covariance and actual joint covariance
$\mu_{\mathcal{K}}$	uniform probability on one labelled feasible configuration space
L_*, J	deterministic proposal majorant and expected proposal-budget bound

Table 1: Notation guide. Reference time is t; the corresponding physical time is Ht.

2 Model and main theorem

2.1 Target chemistry and numerical charge

A finite labelled reaction table consists of m channels $r : y_r \to z_r$ on d species, with $y_r, z_r \in \mathbb{N}^d$ and rational $0 < k_r \le 1$. Complexes are given as expanded lists of molecules. Set $q_r = |y_r|_1$ and $D = \max(1, \max_r |y_r|_1, \max_r |z_r|_1)$. With the ordered-slot convention, the target intensity is

$$k_r(x)_{y_r}, \qquad (x)_y = \prod_i x_i(x_i - 1)\cdots(x_i - y_i + 1). \tag{2.1}$$

There is no factorial division when a reactant is repeated. Labels may be parallel or null; a null label still records a firing time. Empty sums and maxima throughout are zero, with the explicit minimum $D \ge 1$ retained.

The input includes a reverse involution $r \mapsto \bar{r}$, with $y_{\bar{r}} = z_r$, $z_{\bar{r}} = y_r$. A fixed label of the involution is null. It includes rational activities $a_i > 0$ and positive integers ω_i such that, for every label,

$$k_r a^{y_r} = k_{\bar{r}} a^{z_r}, \qquad \omega \cdot y_r = \omega \cdot z_r. \tag{2.2}$$

The initial count $x_0 \in \mathbb{N}^d$ is binary encoded and $N = N_0 = \omega \cdot x_0$. The target chain is nonexplosive because its conserved count set is finite.

The remaining rational input is $\tau > 0$, $0 < \epsilon < 1/8$, $0 < d_0 \le 1$, $\Lambda \ge 1$, $0 < r \le 1/8$, and a full-dimensional convex polytope $\Omega \subset [-1, 1]^3$. It is specified by F nonzero rational facet inequalities, with redundant inequalities allowed, a rational point c, and $0 < \kappa \le 1$ certifying $B_\kappa(c) \subset \Omega$. We write its inequalities as $n_f \cdot x \le b_f$, where n_f is the outward unit normal. These normalized coefficients may be algebraic.

Define the following numerical quantities before making an efficiency claim:

$$\begin{aligned} B &= 1 + \left\lceil \frac{256(N+1)}{\epsilon} \right\rceil, & U &= 1 + B, \\ \Gamma &= 2 + m + \sum_r k_r U^{q_r}, & \Xi &= 2 + \left(2 + \sum_i (a_i + a_i^{-1})\right)^{2D+2}, \end{aligned} \tag{2.3}$$

and

$$\begin{aligned} R_{\mathrm{num}} = 2 + d + m + D + F + \sum_i \omega_i + N + \tau + \tau^{-1} + \epsilon^{-1} + d_0^{-1} + \Lambda + r^{-1} + \kappa^{-1} \\ + \Gamma + \sum_r k_r^{-1} + \Xi. \end{aligned} \tag{2.4}$$

Let L be the complete binary input length. A polynomial in R_{num} can be very large as a function of L. In particular, variable reaction order, activities, time and initial population are paid numerically.

2.2 Particles, reactions and faults

A fine species s carries a nonzero column $P_s \in \mathbb{N}^d$. Put $d_s = |P_s|_1$ and give it radius $\rho_s = \rho d_s$ for one computed rational $\rho > 0$. Fine counts Z decode to PZ. For a current labelled particle list of length n, let $s(i)$ denote species and $r_i = \rho_{s(i)}$. The feasible positions are

$$\mathcal{K} = \{q : n_f \cdot q_i \le b_f - r_i, \quad |q_i - q_j| \ge r_i + r_j\}. \tag{2.5}$$

Labelling is an analytical device, consistent with exchangeability within a species. Actual mass-action counts use ordered distinct slots.

Let $V = \mathrm{Vol}(\Omega)$, $v = (2r)^3$ and

$$K_r(z) = v^{-1}\mathbf{1}_{[-r,r]^3}(z). \tag{2.6}$$

There is no periodic identification. A binary-to-unary channel acts on an ordered pair at x, y with nominal attempt rate $c_e V K_r(y - x)$ and retains the anchor x. Its reverse acts on a parent at x with attempt rate $c_{\bar{e}}$ and draws an independent offset $z \sim K_r$, placing children at $x, x + z$. Unary channels retain the center; binary-to-binary channels retain their designated slots. A global $0 \to 0$ label has its stated rate. A jump succeeds precisely when its proposed new configuration is feasible. An infeasible attempt is recorded as a hidden null. The structural list of such branches is specified in section 3.2; success and rejection branches have independently perturbed intensities. Neither branch is removed because it is unlikely at a particular count or placement.

For each species, the unknown nominal diffusivity is a deterministic real symmetric matrix $D_s(t, x)$ satisfying

$$d_0 I \le D_s(t, x) \le I. \tag{2.7}$$

It is C^3 in space up to the boundary, its spatial derivatives of orders $1, 2, 3$ have operator norm at most Λ, and D_s and its spatial derivatives through order two are Lipschitz in time with constant at most Λ. These bounds hold for all physical times. Set

$$g_s = \operatorname{div} D_s, \qquad A_0(t, q) = \operatorname{diag}(D_{s(i)}(t, q_i)).$$

The divergence drift g_s is part of the model, rather than an imposed control. Between chemical events the positions obey

$$\begin{aligned} dQ_i = {} & (g_{s(i)}(t, Q_i) + v_i)\, dt + [\sqrt{2A}\, dW]_i - \sum_f D_{s(i)}(t, Q_i) n_f\, dL_{if} \\ & + \sum_{j \ne i} D_{s(i)}(t, Q_i) e_{ij}\, dL_{ij}, \qquad e_{ij} = \frac{Q_i - Q_j}{|Q_i - Q_j|}. \end{aligned} \tag{2.8}$$

Each L_{if} is continuous nondecreasing and grows only at the indicated wall contact. The common $L_{ij} = L_{ji}$ grows only at pair contact. They are reset as lifetime clocks when the particle list changes.

For a computed $0 < \delta \le 1/4$, allowed faults satisfy, predictably,

$$|v_i| \le \delta, \qquad A = A^\top, \quad \|A - A_0\|_{\mathrm{op}} \le \delta d_0, \tag{2.9}$$

and multiply every success, rejection and other chemical branch separately by a factor in $[1 - \delta, 1 + \delta]$. The joint matrix A need not be block diagonal. Faults may depend on the full continuous history, hidden and null events, contact clocks and external randomness, but not on future innovations. Reflection in (2.8) remains the *nominal block-conormal* reflection; it is not replaced by conormal reflection for A.

An *admitted realization* is a process satisfying these equations and clock conditions with Brownian and reaction innovations relative to its filtration. The theorem covers every such realization and every admitted clock decomposition. We also prove existence and uniqueness of the nominal position process, including from touching configurations. Existence for an arbitrary pathological feedback specification is not an additional assertion.

2.3 Path observation and theorem

The retained record is the initial decoded count together with the ordered list (t_j, r_j, x_j) of original-labelled firings and post-firing decoded counts. The exact real time coordinates are part of this measurable record. Original null and parallel labels are kept. Positions, private reactions and steric rejections are hidden. For laws ν, ν' on this record space, $d_{\mathrm{TV}}(\nu, \nu') = \sup_E |\nu(E) - \nu'(E)|$.

Theorem 2.1 (Finite-radius spatial compilation)**.** *There are computable universal constants C, p and a deterministic compiler with the following properties for every input in section 2. It emits an elementary labelled network, payload matrix P equal to the identity on original species, positive rational activities $\widetilde{a}_s$ with $\widetilde{a}_i = a_i$, positive rational coefficients $c_e \le 1$, one positive rational radius scale ρ, a rational slowdown $H \ge 1$ and a rational tolerance $\delta > 0$.*

The species, base-channel and payload structure depends only on the reaction list and reverse involution. Every column of P is nonzero and

$$\rho(1 + N + P_{\max}) \le \frac{\min(\kappa, r)}{2^{16}}, \qquad P_{\max} = \max_s |P_s|_1. \tag{2.10}$$

Every actual reverse pair satisfies $c_e \widetilde{a}^{\alpha_e} = c_{\bar{e}} \widetilde{a}^{\beta_e}$, with the designated reverse labels. These are nominal algebraic identities.

For all finite fine counts u, w,

$$u \rightsquigarrow w \quad \Longleftrightarrow \quad Pu \rightsquigarrow Pw, \tag{2.11}$$

and communicating classes are the corresponding preimages. On the physical cap $\omega \cdot Pz \le N$, every count has strictly feasible configurations, and every finite allowed count path has positive-probability spatial realizations from every feasible initial placement.

Initialize exactly x_0 original particles at any feasible deterministic positions and no auxiliaries. For every profile satisfying (2.7) and every admitted fault realization,

$$d_{\mathrm{TV}}\big(\mathcal{L}(\text{observed physical path at times } Ht,\ 0 \le t \le \tau), \mathcal{L}(X_{[0,\tau]})\big) \le \epsilon. \tag{2.12}$$

No initial interval or internal waiting time is deleted.

The nominal diffusion and successful jumps have a common restricted Poisson reference measure, as specified in proposition 3.2. For all actual physical times $s \ge 0$, $|Z_s|_1 \le N$ pathwise, the process is nonexplosive, and all the resource bounds of theorem 7.1 hold. Compiler time, memory, output length, $H, H\tau, \rho^{-1}, \delta^{-1}$, positive coefficient and activity inverses, and the stated horizon resource bounds are at most

$$C(1 + L + R_{\mathrm{num}})^p. \tag{2.13}$$

The compiler does not query the unknown profiles or compute a configuration-space normalizer, a PDE, a metric projection or a path-law sampler.

The abstract graph assertion (2.11) concerns finite integer counts without a packing interpretation. Its spatial assertion is restricted to the conserved cap. In particular, the theorem does not claim that an unbounded number of positive balls fits in a bounded vessel.

3 Elementary chemistry and exact spatial balance

3.1 The precise kinetic interface

The first compiler operates on counts. We record its contract so that the spatial proof has a public, noncircular dependency.

Proposition 3.1 (Paired kinetic interface, [6, Theorem 9.1]). *For the chemical data of section 2, duration τ and accuracy $\epsilon/8$, a deterministic algorithm produces $d+J_0$ species and $2J_0+m$ channels, where $J_0 = \sum_r q_r$, a nonzero-column integer payload matrix P with $P_{\max} \le D$, activities $\widetilde{a}_s$, normalized coefficients $0 < b_e \le 1$, a clock $H_{\mathrm{kin}} \ge 1$ and tolerance $\delta_{\mathrm{kin}} > 0$. Base channels are $2 \leftrightarrow 1$, unary-to-unary, and global $0 \to 0$ channels. They have the graph and label properties in theorem 2.1, conserve $\omega \cdot PZ$, and are exactly detailed balanced at $\widetilde{a}$. From $(x_0, 0)$ their complete decoded marked path on $[0, H_{\mathrm{kin}}\tau]$, rescaled by H_{kin}, is within $\epsilon/8$ of the target, under every full-history predictable relative chemical fault of size at most δ_{kin}.*

With W_b as in that theorem, $W_b \le 8R_{\mathrm{num}}$. Every numerical output, including H_{kin}, $H_{\mathrm{kin}}\tau$, $\delta_{\mathrm{kin}}^{-1}$, $\max_e b_e^{-1}$ and $\max_s(\widetilde{a}_s + \widetilde{a}_s^{-1})$, is bounded by $2^{1024}W_b^{128}$. Its bit work, memory and output length are at most $2^{1024}(1+L)^{128}$.

The cutoff used by the cited theorem is $1 + \lceil 32(N+1)/(\epsilon/8) \rceil$, exactly B in (2.3). Thus its variable-order activity charge and Γ are the same quantities, rather than estimates at an uncharged larger cutoff. All its other hypotheses are precisely (2.2). The theorem is proved independently of any spatial construction.

For orientation, its paired construction can be described explicitly. For every directed source list $y_r = (i_1, \dots, i_{q_r})$, make private prefixes $S_{r,j}$ of payload $\sum_{h \le j} e_{i_h}$, $2 \le j \le q_r$, and a pending species C_r of payload y_r when $q_r > 0$. The chains are

$$X_{i_1} + X_{i_2} \rightleftarrows S_{r,2}, \quad S_{r,j-1} + X_{i_j} \rightleftarrows S_{r,j}, \quad S_{r,q_r} \rightleftarrows C_r.$$

For a unary source the chain is $X_{i_1} \rightleftarrows C_r$; an empty conservative source has an empty product and uses a global null channel. Central observed channels are $C_r \to C_{\bar{r}}$ with label r. Release to originals connects every fibre to its canonical state. Hence every fine transition decodes correctly, and every target edge lifts; this proves the exact graph assertion without count enumeration.

To see why the paired lift is balanced, assign relative weights $w(S_{r,j}) = \vartheta U^{-j}$ and $w(C_r) = \vartheta U^{-q_r}$, and absolute activities $\widetilde{a}_s = w(s)a^{P_s}$, with original relative weight one. First binding/release coefficients before normalization are $K\vartheta/U^2, K$; subsequent ones are $K/U, K$; full-prefix/pending coefficients are K, K; unary entry/return coefficients are $K\vartheta/U, K$. A central coefficient is $Qk_rU^{q_r}$. Each central balance identity reduces to $Q\vartheta k_r a^{y_r} = Q\vartheta k_{\bar{r}} a^{z_r}$. The quantitative fast-fibre and marked-filter argument determining ϑ, K, Q and the common normalization is the proof of the cited theorem. In particular, it includes every post-event layer and original null mark. Reversibility is obtained by this paired construction, rather than by restricting an irreversible compiler to a conservative table.

3.2 Local jumps and structural rejection

For a base channel, use the placement rule of section 2. Add one hidden rejection branch precisely in the following cases:

Base type	Condition for a rejection branch
$1 \to 2$	Always
$1 \to 1$	The retained slot grows
$2 \to 2$	At least one retained slot grows
$2 \to 1$	The anchored product grows beyond the anchored reactant

The branch retains all reactant slots and their species, and fires only on the failure set of its base placement. Thus a binary rejection is a binary null channel with the *inherited failure kernel*, not an

unrestricted identity-placement channel. The base network in proposition 3.1 needs binary-to-binary channels only for these null branches.

If no retained slot grows and no new slot is added, feasibility is preserved by removal and shrinking, proving that no rejection branch is possible. A $1 \to 2$ release fails for sufficiently small offsets. A growing slot can be placed near a supporting face so that its larger replacement fails. The other particles can be parked in the inball. These are open sets of placements and, where needed, offsets. A supporting face nearest c gives a usable patch without a lower bound on an individual facet's area. Tiny radii relative to κ leave the patch and parking region separated. Consequently the rejection list is structural and independent of population or numerical geometry. A genuine identity null has identical slot radii and always succeeds, retaining its original label.

At zero faults, the success and failure coefficients are the same. Under faults their multipliers are independent. For example, if $w(q)$ is a release-success probability at the current placement, its actual combined attempt rate is

$$c\{\alpha_{\mathrm{suc}} w(q) + \alpha_{\mathrm{rej}}(1 - w(q))\}.$$

This need not be constant. The proof below uses a majorizing proposal process, rather than treating this rate as an unperturbed Poisson clock.

3.3 The encounter normalizer

The point-particle normalization is the rational number

$$q = \frac{1}{V} \int_{\Omega^2} K_r(y - x)\, dx\, dy, \qquad \frac{\kappa^6}{4096} \le q \le 1. \tag{3.1}$$

The upper bound follows by integrating the probability kernel. For the lower bound, put $s = \min(\kappa/2, r)$ and integrate over $x \in B_{\kappa/2}(c)$, $z \in B_s(0)$, $y = x + z$. These pairs lie in Ω^2 and in the cube kernel. Their contribution is at least $\pi^2 \kappa^3 \min(\kappa/(2r), 1)^3/288$. Using $r \le 1/8$ and $\kappa \le 1$ gives the displayed conservative lower bound. The double integral is a six-dimensional rational polytope volume divided by Vv. Its exact finite-bit calculation is given in appendix C. No hard-ball configuration volume is calculated.

After an extra integer slowdown h, set

$$c_e = \begin{cases} b_e/(hq), & e \text{ belongs to a binding/release pair,} \\ b_e/h, & e \text{ is unary or global null.} \end{cases} \tag{3.2}$$

Rejection branches inherit the corresponding base coefficient. Reverse pairs receive the same factor, preserving all activity identities exactly.

Proposition 3.2 (Exact nominal flux)**.** *At zero faults the successful spatial jump kernels satisfy detailed balance with the Poisson point measure of intensities $\widetilde{a}_s/V$, restricted to feasible configurations and to any invariant chemical class or conserved cap. The fixed-count nominal diffusion has the instantaneous symmetric form*

$$\mathcal{E}_t(f, g) = \int_{\mathcal{K}} \nabla f^{\top} A_0(t, q) \nabla g \, d\mu_{\mathcal{K}}, \qquad \mathrm{Dom}\, \mathcal{E}_t = H^1(\mathcal{K}). \tag{3.3}$$

Uniform measure $\mu_{\mathcal{K}}$ is common to these forms. A restricted measure is normalized only when it has positive mass; every count beneath our cap has strictly feasible configurations. These assertions do not state reversibility of a time-ordered evolution or of a faulted marginal.

Proof. Fix the untouched configuration η and a reversible binding $A + B \rightleftarrows C$. The two successful Palm fluxes have the same geometric measure

$$K_r(y-x)\mathbf{1}_{\mathcal{K}}(\eta + A_x + B_y)\mathbf{1}_{\mathcal{K}}(\eta + C_x)\,dx\,dy.$$

Their scalar factors are, respectively,

$$\frac{c_e \widetilde{a}_A \widetilde{a}_B}{V} \quad \text{and} \quad \frac{c_{\bar{e}} \widetilde{a}_C}{V}.$$

These are equal. Ordered identical-reactant slots give the factorial Palm factor $\widetilde{a}_A^2$, with no division by two. This also follows directly by cancellation of the count factorial with the falling factorial. Unary retained-center fluxes have the same feasible source/target indicator and equal activity factors. Rejections and global nulls are diagonal configuration fluxes. Restricting both endpoints to an invariant set preserves each equality. No normalizer enters this calculation.

The nominal generator is $\operatorname{div}(A_0\nabla)$ because the intrinsic drift is $\operatorname{div} A_0$. Integration by parts gives (3.3) with conormal Neumann boundary. The construction from all boundary starting points and the identification of this form are proved in appendix A.3. In particular, constants are preserved and the uniform measure is invariant at every time. □

For a fine count z, the reference count weight is proportional to

$$\frac{\widetilde{a}^z}{z!}\,\frac{\operatorname{Vol}(\mathcal{K}_z)}{V^{|z|_1}}. \tag{3.4}$$

The excluded-volume factor depends on z. In addition the kinetic decoder has its own fibre partition correction. Neither correction is suppressed; the decoded equilibrium is not asserted to be the target's unconditional Poisson count law.

Example 3.3 (A boundary rejection in $2A \rightleftarrows B$)**.** Take both target rates and activities equal to one and weights $\omega_A = 1$, $\omega_B = 2$. The forward prefix and pending species carry $2A$; the reverse pending species carries B. Their radii are $2\rho, 2\rho, \rho$, respectively: radii use unweighted payload size, not conserved mass. An elementary binding $A + A \to S$ anchors the product of radius 2ρ at the first center. It may fail near a wall even when the two reactants fit. Its reverse samples the offset in the same cube and checks both offspring. The two successful fluxes contain exactly the same pair of feasibility indicators, so rejecting one orientation without the corresponding indicator in the other would break balance. The failure is a hidden attempt; it is not an observed $2A \to B$ event.

For the unit cube $[-1/2, 1/2]^3$, the point-particle normalizer is explicitly $q = (1 - r/2)^3$. It is neither the release-success probability at every parent position nor the hard-ball count normalizer in (3.4).

4 Dilute hard-ball geometry

All fixed-count arguments use only $n \le N$, $r_i = \rho d_i$, $d_i \ge 1$ and $\sum_i d_i \le N$. Write the inward gaps

$$g_{if} = b_f - n_f \cdot q_i - r_i, \qquad g_{ij} = |q_i - q_j| - r_i - r_j, \tag{4.1}$$

whose gradients have norms one and $\sqrt{2}$. There are at most $J_c = nF + n(n-1)/2$ constraints. The empty configuration requires no estimates.

4.1 A common inward direction

Lemma 4.1 (Contact cone)**.** *Under $\rho(1+N) \le \kappa/2^{16}$, at every feasible configuration there is a unit vector v with*

$$\nabla g_j \cdot v \ge \alpha, \qquad \alpha = \frac{\kappa}{2^{10}(1+N)^2}, \qquad g_j \le \rho. \tag{4.2}$$

For every nonnegative combination of active conormals,

$$\left| \sum_j \lambda_j A_0 \nabla g_j \right| \ge d_0 \alpha \sum_j \lambda_j. \tag{4.3}$$

Proof. Join particles when their pair gap is at most ρ. In each connected cluster C, put $R_C = \sum_{i \in C} r_i$ and let z_C be the arithmetic mean of its centers. Along a simple connecting path, each edge has length at most $r_i + r_j + \rho$; thus the cluster diameter is at most $3R_C$. Use the unnormalized direction

$$v_i = \frac{16 R_C}{\kappa}(c - z_C) + (q_i - z_C), \qquad i \in C.$$

For a near pair its gap derivative is $|q_i - q_j| \ge 2\rho$. For a near wall, $n_f \cdot (c - z_C) \le -\kappa + 5R_C \le -\kappa/2$, so $n_f \cdot v_i \le -8R_C + 3R_C \le -5\rho$. Finally $|v| \le 60\rho N^{3/2}/\kappa$. Normalization gives (4.2) with room to spare. Pairing the conormal sum with $A_0^{-1}v$, whose norm is at most d_0^{-1}, gives (4.3). □

The angle α is independent of ρ. The neighborhood on which it works shrinks with ρ, which will determine a short time scale. In particular a positive circulation of contact clocks cannot hide a large unsigned push behind a zero resultant.

4.2 Poincaré by buffering, separation and coordinate paths

Theorem 4.2 (Dilute hard-ball Poincaré inequality)**.** *Let $\rho = a^2$ for a positive rational a satisfying*

$$a \le 2^{-30}(2 + N + F + \kappa^{-1})^{-20}. \tag{4.4}$$

For every count beneath the cap and every $f \in H^1(\mathcal{K})$,

$$\mathrm{Var}_{\mu_\mathcal{K}} f \le C_P \int_\mathcal{K} |\nabla f|^2 \, d\mu_\mathcal{K}, \qquad C_P = 2^{20}(1+N). \tag{4.5}$$

The obstacle to a direct product comparison is geometric. A line joining two feasible configurations can cross a collision surface; freezing all but one particle can even create a cage. We first move each endpoint a little, with controlled density distortion. The moved endpoints are well separated. A randomly chosen intermediate configuration then supplies two admissible one-particle-at-a-time paths with uniformly positive probability.

Proof. We prove the inequality for smooth functions, then extend it.

Step 1: create a tiny buffer. Choose $0 \le \chi \le 1$, equal to one on $(-\infty, 1/4]$, zero on $[1/2, \infty)$ and Lipschitz with constant eight. Extend pair gradients by replacing $(q_i - q_j)/|q_i - q_j|$ with $(q_i - q_j)/\max(|q_i - q_j|, \rho)$; denote the constraint vectors by b_j. They equal ∇g_j on $\mathcal{K}$. Let $V(q)$ minimize

$$\frac{1}{2}|v|^2 + 8\alpha^{-2} \sum_j \chi(g_j(q)/\rho)[1 - b_j(q) \cdot v]_+^2, \tag{4.6}$$

and clip it to $|v| \le \alpha^{-1}$. On $\mathcal{K}$ the direction in lemma 4.1, divided by α, makes every active penalty zero. Comparison with it proves that clipping has no effect and $|V| \le \alpha^{-1}$. For $g_j \le \rho/4$, comparison of that one penalty gives $\nabla g_j \cdot V \ge 3/4$.

Strong convexity of (4.6) gives the global safe estimate

$$\mathrm{Lip}(V) \le L_V = \frac{2^{16}(1+J_c)^2}{\alpha^3 \rho}. \tag{4.7}$$

For completeness, comparison with $v = 0$ first bounds the untruncated minimizer by $4\sqrt{J_c}/\alpha$. Subtract the two minimizer equations, pair with their difference, and use strong monotonicity on the left. The coefficient differences on the right are bounded using $\mathrm{Lip}(b_j) \le 4/\rho$, $\mathrm{Lip}(\chi \circ (g_j/\rho)) \le 16/\rho$ and this minimizer bound. Their sum is at most the right side of (4.7) times the displacement. Radial clipping is nonexpansive.

Set

$$\eta = 64\rho^3(1+N)^5\kappa^{-2}, \tag{4.8}$$

and let B_s be the flow of V for time $2\eta s$, $0 \le s \le 1$. The positive derivative of every small gap proves forward invariance and $B_1\mathcal{K} \subset \mathcal{K}_\eta := \{g_j \ge \eta\}$. The flow is injective, with inverse Lipschitz constant $e^{2\eta L_V}$. Hence its volume distortion is at most $e^{6n\eta L_V}$. If $G_0 = 2 + N + F + \kappa^{-1}$, then

$$6n\eta L_V \le 2^{55}\rho^2 G_0^{21} < \log 2$$

by (4.4). Thus $(B_s)_{\#}\mu_{\mathcal{K}} \le 2\mu_{\mathcal{K}}$, and integration along the flow gives

$$\int |f - f \circ B_1|^2 \, d\mu_{\mathcal{K}} \le 8\eta^2\alpha^{-2} \int |\nabla f|^2 \, d\mu_{\mathcal{K}}. \tag{4.9}$$

Step 2: separate by rank without a factorial penalty. For buffered x, sort the first coordinates and put $k_i = \mathrm{rank}(x_{i1}) - (n+1)/2$, breaking ties by label. Define

$$\theta = 4an/\kappa, \qquad S_s(x)_i = (1 - s\theta)x_i + s\theta c + sak_i e_1. \tag{4.10}$$

For a wall, its gap is at least $(1 - s\theta)\eta + s\theta(\kappa - r_i) - sa|k_i| \ge \eta$. For a pair, put $z = x_i - x_j$, $k = k_i - k_j$. Sorting gives $z_1 k \ge 0$ and $|k| \ge 1$, so

$$\begin{aligned} |S_s(x)_i - S_s(x)_j|^2 &\ge (1 - s\theta)^2|z|^2 + s^2a^2k^2 \\ &\ge \frac{|z|^2}{1 + \theta^2|z|^2/(a^2k^2)}. \end{aligned} \tag{4.11}$$

The last bound is obtained by minimizing the quadratic over all real s. It increases with $|z|^2$. With $R = r_i + r_j \le \rho N$ and $|z| \ge R + \eta$, its possible loss in length is at most $8n^2(R+\eta)^3/\kappa^2$. Condition (4.4) gives $\eta \le \rho N/8$ when $N \ge 1$, and the loss is less than $\eta/4$ by (4.8). Thus the whole path is strictly pair feasible. Its endpoint has first-coordinate separations at least a.

On each rank chamber the Jacobian is $(1 - s\theta)^{3n} \ge 1/2$. The output ordering recovers the input chamber. Distinct chamber images are disjoint: summing their density contributions costs no factor $n!$. Tie sets and their preimages under the bi-Lipschitz buffer have measure zero. The squared speed is at most $A_{\mathrm{rank}}^2 = 2^{10}\rho n^3\kappa^{-2}$. For $T = S_1B_1$ and $\lambda = T_{\#}\mu_{\mathcal{K}}$, we have

$$\lambda \le 4\mu_{\mathcal{K}}, \tag{4.12}$$

$$\int |f - f \circ T|^2 \, d\mu_{\mathcal{K}} \le (16\eta^2\alpha^{-2} + 8A_{\mathrm{rank}}^2) \int |\nabla f|^2 \, d\mu_{\mathcal{K}}. \tag{4.13}$$

Step 3: compare through a common intermediate point. Let $\Omega_i = \{x : n_f \cdot x \le b_f - r_i\}$ and $\nu = \bigotimes_i \mathrm{Unif}(\Omega_i)$. Then $\mu_{\mathcal{K}} = \nu(\cdot \mid \mathcal{K})$. Each Ω_i contains $B_{\kappa/2}(c)$. A union bound for pair exclusion, using the volume of a three-dimensional ball, gives $\nu(\mathcal{K}) \ge 1/2$ under (4.4).

Starting at an a-separated configuration u, draw $z \sim \nu$ and move particle 1 from u_1 to z_1, then particle 2, and so on. Walls remain feasible by convexity. For an unmoved obstacle u_j, $j > i$, endpoints z_i producing a collision with its exclusion ball occupy a solid-angle cone of probability at most $C(r_i + r_j)^2/(\kappa^3 a^2)$: the cone angle is at most a constant times $(r_i + r_j)/|u_i - u_j|$, and its length is bounded by the vessel diameter. For a moved obstacle z_j, condition on z_i. Its bad locations occupy a tube about $[u_i, z_i]$, of probability at most $C(r_i + r_j)^2/\kappa^3$. Because

$$\sum_{i<j}(d_i + d_j)^2 = (n-2)\sum_i d_i^2 + \Big(\sum_i d_i\Big)^2 \le N^3,$$

the probability that the entire path fails is at most

$$\beta = 2^9 \kappa^{-3} \rho^2 N^3 (1 + a^{-2}) \le \tfrac{1}{4}. \tag{4.14}$$

For any two separated endpoints u, w, a common z supports both paths with probability at least $1 - 2\beta \ge 1/2$.

To bound their energies, during segment i the current configuration is

$$(z_1, \dots, z_{i-1}, (1-s)u_i + sz_i, u_{i+1}, \dots, u_n).$$

For $s \le 1/2$ change variables from u_i to the current coordinate; for $s \ge 1/2$ change from z_i. Each Jacobian is at most $2^3 = 8$. All other unused coordinates integrate out. Using $\lambda \le 4\nu/\nu(\mathcal{K})$ and retaining the valid-path indicator, the remaining energy integral is over $\mathcal{K}$; its factor $\nu(\mathcal{K})$ cancels. Segment lengths squared are at most 12. Cauchy–Schwarz over the n segments gives integrated energy at most $384n \int |\nabla f|^2 \, d\mu_{\mathcal{K}}$ for one endpoint. The two paths, the inequality $|a+b|^2 \le 2a^2 + 2b^2$, and the validity normalization at most two yield

$$\mathrm{Var}_\lambda f \le 1536n \int |\nabla f|^2 \, d\mu_{\mathcal{K}}. \tag{4.15}$$

Only one three-dimensional coordinate was interpolated at a time.

Choose the mean of f under λ as a comparison constant and use (4.13) and (4.15). We obtain

$$\mathrm{Var}_{\mu_{\mathcal{K}}} f \le (32\eta^2\alpha^{-2} + 16A_{\mathrm{rank}}^2 + 3072n) \int |\nabla f|^2 \, d\mu_{\mathcal{K}}.$$

Condition (4.4) absorbs the first two terms into $2^{20}(1+N)$. Finally, the cone condition makes $\mathcal{K}$ a bounded Lipschitz domain. Sobolev extension and smooth density extend the inequality to $H^1(\mathcal{K})$. No quantitative extension constant enters the proved estimate. □

4.3 Feasibility and support

The preceding construction proves more than an integral inequality. Every feasible point can be buffered and rank-separated by a continuous feasible path, strictly feasible after the initial buffer. Any two separated points share valid intermediate points of positive volume. Almost every such intermediate avoids tangencies, giving strictly feasible coordinate segments. Thus every feasible starting configuration connects to any prescribed interior neighborhood. Strict configurations also exist directly by putting the capped particles in separated slots along a line in $B_\kappa(c)$.

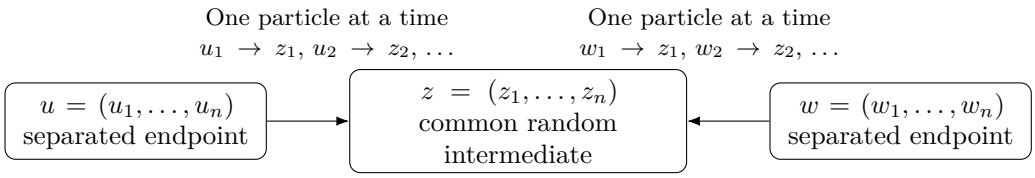


Figure 2: The path comparison in theorem 4.2. The endpoint rank deformation also has disjoint chamber images, avoiding a separate $n!$ density loss. These are integration paths, not particle controls.

Uniform ellipticity and a bounded-drift Girsanov guide along a finite piecewise smooth interior path give positive probability of following a small tube. At an initial contact, use the inward buffer and the short reflection estimate in lemma 5.1; then use interior segments. Equivalently the metric error estimate in appendix A.2, with the guide drift, gives positive endpoint probability on sufficiently short segments. Here are the quantitative facts needed for this qualitative use. Subdivide the guide until each segment has displacement small relative to the local cone radius. Traverse one segment in time u with its prescribed guide drift. The core argument bounds the conditional expected clock by $C\sqrt{u}$ plus a polynomial constant times that segment's length. Further subdivision makes the latter small enough for an exponential clock bound. Applying the metric comparison to the actual motion and the deterministic feasible guide then gives mean squared endpoint error at most $C'u$. Choose u smaller than a fixed fraction of the squared clearance of the desired endpoint neighborhood. Markov's inequality gives positive probability of entering it. The guide has finite bounded speed on each chosen segment. Uniform ellipticity makes its Girsanov density strictly positive with finite finite-horizon exponential bound. Thus the same event has positive probability under the original drift. This argument applies to the initial inward segment as well as strict interior segments. This qualitative argument is valid under adapted covariance faults as well: its inverse covariance is uniformly bounded, and no derivative of the fault is used. An independent proposal-free gap has positive probability. At suitable interior placements every enabled successful channel has positive intensity and a positive-volume set of successful offsets. The failure placements in section 3.2 give the same conclusion for each listed rejection. Iteration proves finite-path support. No polynomial lower bound on these individual path probabilities is claimed or used in the accuracy estimate.

5 Unknown transport and conditional mixing

The nominal reference measure is known as a mathematical object: it is uniform on $\mathcal{K}$. Its normalizing volume is not known to the compiler. Unknown diffusivity affects the rate of relaxation, but not this measure. To exploit this fact we need estimates uniform over counts, initial contacts, time origins and permitted coefficient profiles.

We give an intentionally coarse explicit ledger. Define

$$Z_* = 2 + N + F + \kappa^{-1} + \rho^{-1} + d_0^{-1} + \Lambda + r^{-1}, \qquad \mathsf{M} = 2^{1024} Z_*^{1024}. \tag{5.1}$$

The large fixed powers organize resource dependencies; they are not proposed performance estimates.

Set

$$\ell = \mathsf{M}^{-16}, \quad G = \mathsf{M}^{12}, \quad b(g) = \min\{2, \mathsf{M}^{16} e^{-g/\mathsf{M}^4}\}, \quad I = \mathsf{M}^6, \quad W = \mathsf{M}^{64}. \tag{5.2}$$

Only I, W are evaluated for the final parameter choices. The exponential envelope is a proof device; no transcendental number is output.

Lemma 5.1 (Local reflected services)**.** *For every fixed capped count, the nominal equation has a unique position process from every $q \in \mathcal{K}$, with form (3.3). Its evolution is denoted $P^0_{s,t}$. Let $\mathcal{L}$ be the sum of all wall clocks and unordered pair clocks on a chemical-free interval. Uniformly over full past histories and all admitted realizations:*

(i) For $0 < u \le \ell$, $\mathbb{E}[\Delta\mathcal{L} \mid \mathcal{F}_s] \le \mathsf{M}\sqrt{u}$. On such windows, $\mathbb{E}[e^{8\mathsf{M}\Delta\mathcal{L}} \mid \mathcal{F}_s] \le 2$.

(ii) Couple the actual and nominal motion from the same point with the same Brownian innovation. For $u \le \ell$, $\mathbb{E}[|Q_{s+u} - Q^0_{s+u}| \mid \mathcal{F}_s] \le \mathsf{M}^4\delta$.

(iii) For $u \in [\ell/2, \ell]$ and bounded measurable f, $P^0_{s,s+u}f$ is Euclidean Lipschitz on $\mathcal{K}$ with constant at most $G\,\|f\|_\infty$. The same bound holds for the gradient norm and boundary values.

These constants do not depend on derivatives of the actual faults.

The proof occupies appendix A. Its main device is the distance to a slightly tightened configuration core, measured in the nominal inverse covariance metric. Each individual contact lowers this potential. This pays unsigned clocks even at corners, then supplies the exponential moments needed for comparison and smooth-boundary approximation. The smoothing import is applied only on smooth approximants, with polynomial curvature and local-time constants checked uniformly before taking the corner limit.

Lemma 5.2 (Uniform nonautonomous mixing)**.** *For all $s, g \ge 0$ and $\|f\|_\infty \le 1$,*

$$\sup_{q\in\mathcal{K}} |P^0_{s,s+g}f(q) - \mu_{\mathcal{K}}f| \le b(g), \qquad \int_0^\infty b(g)\,dg \le I. \tag{5.3}$$

Proof. Constants are preserved and all instantaneous forms have the same invariant measure. Their energy identity and theorem 4.2 give

$$\left\| P^0_{s,s+g}f - \mu_{\mathcal{K}}f \right\|_2 \le e^{-d_0 g/C_P} \left\| f - \mu_{\mathcal{K}}f \right\|_2 . \tag{5.4}$$

One may see this by reversing the time coordinate in the backward equation and differentiating the squared norm. No commutation or self-adjointness of the full evolution is required.

Let $m_* = 3n$. The contact cone and the pair Hessian bound give, on a radius at least M^{-1},

$$\mu_{\mathcal{K}}(B(q,u) \cap \mathcal{K}) \ge \left(\frac{\alpha u}{2^{12}\sqrt{m_*}} \right)^{m_*} . \tag{5.5}$$

Indeed move a distance $u/2$ in a common inward direction and put a ball of radius $\alpha u/16$ about that point. For sufficiently small u the quadratic gap loss is smaller than its linear gain; initially far constraints remain far. The volume of an m_*-ball is at least $(c/\sqrt{m_*})^{m_*}$, while $\mathrm{Vol}(\mathcal{K}) \le 8^n$. The local geometric constants are dominated by (5.1).

For $g \ge \ell$, smoothing the first short interval bounds the local Lipschitz constant by G. If the centered function has maximum absolute value M_f, it stays above $M_f/2$ in absolute value in a feasible ball of radius $M_f/(2G)$. This radius is within the range of (5.5). Consequently

$$M_f^{m_*+2} \le 4 \left(\frac{2^{13} G\sqrt{m_*}}{\alpha} \right)^{m_*} \left\| P^0_{s,s+g}f - \mu_{\mathcal{K}}f \right\|_2^2 .$$

Taking the (m_*+2)th root makes the prefactor polynomial. By (5.1), m_*+2, C_P and d_0^{-1} are each bounded by M; the resulting bound is at most $\mathsf{M}^{16}e^{-g/\mathsf{M}^4}$. Short times are covered by two. Integration gives at most $\mathsf{M}^4(2+32\log\mathsf{M}) \le \mathsf{M}^6$. For $n=0$ the centered function is zero. □

The small-ball volume has a dimension-dependent power. It is used only before taking a dimension-dependent root; it never becomes a numerical charge raised to the population. In particular we do not use a high-dimensional pointwise heat-density bound.

5.1 Conditional channel observables

A normalized channel observable is the success or failure predicate of a specified local placement, with its cube-contact factor, scaled to lie in $[0,1]$. For release, integrate the predicate over the independent offset. The only discontinuities are cube faces, shifted walls and pair-distance surfaces involving an affected slot. For every such f and $0 < w \le 1/2$ there are upper and lower functions satisfying

$$0 \le f_w^- \le f \le f_w^+ \le 1, \quad \mathrm{Lip}(f_w^\pm) \le \mathsf{M}/w, \quad \mu_{\mathcal{K}}(f_w^+ - f_w^-) \le \mathsf{M}w. \tag{5.6}$$

For products of predicates, replace each step by a linear ramp of width w and bound the product difference by the sum of the individual ramp differences. Pair distances and wall distances are uniformly Lipschitz. The number of affected constraints is polynomial in $N+F$. Each tube volume is three-dimensional: wall and cube slabs have volume $O(w)$, and a distance-shell about a ball has volume $O(w(R+w)^2)$. Current feasible-uniform measure is dominated by twice product-uniform measure under (4.4); a release adds the density factor v^{-1}. These observations prove (5.6), including its stated constants, as detailed in appendix B.1.

Proposition 5.3 (Conditional occupation)**.** *On a chemical-free gap of length g, starting at any feasible configuration, every normalized channel observable satisfies*

$$|\mathbb{E}[f(Q_{s+g}) \mid \textit{full past}] - \mu_{\mathcal{K}} f| \le b(g) + W\delta^{2/9}. \tag{5.7}$$

The same statement holds conditioned on the next independent proposal gap. The motion during the gap may depend on its entire history.

Proof. For $g < 2\ell$ the trivial bound is covered by $b(g) = 2$. Otherwise split the interval into subintervals of length in $[\ell/2, \ell]$ and telescope the actual conditional transitions against the nominal future functions. For a terminal ramp $f_w^\pm$, all but the last term have gradient at most $Gb(u)$, where u is the remaining time after an initial smoothing subinterval. By lemma 5.1(ii), their sum is at most

$$\mathsf{M}^4 G\delta(4 + 4I/\ell) \le \mathsf{M}^{36}\delta.$$

The last term is at most $\mathsf{M}^5\delta/w$. Nominal mixing contributes $b(g)$ and the stationary sandwich contributes $\mathsf{M}w$. This is conditional telescoping, so it assumes neither a Markov fault policy nor independence between actual particles. Taking $w = \sqrt{\delta}$ gives a bound at most $b(g) + \mathsf{M}^{40}\sqrt{\delta}$, which is bounded by (5.7). Larger ramp widths can be covered by the trivial bound; here $\delta \le 1/4$. Conditioning on the next independent proposal gap leaves Brownian innovations and these conditional estimates unchanged. □

Thus robustness is proved through local comparisons and a summable nominal future, rather than a synchronous coupling run for the entire physical storage time. This is what permits a polynomially small fault tolerance without an exponentially accurate simulation of the transport.

6 Comparison of complete timed paths

6.1 Stationary bias and a constant proposal majorant

Let M count all emitted directed branches, including structural rejections, and let $T = H_{\mathrm{kin}}\tau$. Put

$$L_* = \frac{2^{10}(1+M)(1+N)^2(1+8/v)}{q}, \qquad J = 1 + L_*T, \qquad B_* = 2^{10}L_*. \tag{6.1}$$

These quantities are fixed before the radius. At every actual history, including independent faults on success and rejection, total attempt intensity is at most L_*/h. This follows by summing at most $N(N-1)$ ordered pair slots per binary channel, N slots per unary channel, and one per global channel, using $V \le 8$, $K_r \le 1/v$ and $1+\delta \le 5/4$.

Define

$$\zeta_N(\rho) = 3\rho N/\kappa + \rho^3 N^4/\kappa^3. \tag{6.2}$$

For independent uniform centers, the wall layer for slot i has probability at most $3r_i/\kappa$, by contraction toward c through factor r_i/κ. Summing gives $3\rho N/\kappa$. Pair exclusion is bounded by the sum of ball volumes, giving the second term up to a fixed factor. The tiny-radius condition ensures total failure probability below $1/2$. Conditioning on feasibility changes bounded averages by at most a fixed multiple of this probability. The same argument includes the new and removed slots of a reaction, whose combined payload is at most $2N$. Factors from ordered tuples, V/v, release offsets and q^{-1} are paid by B_*.

More explicitly, if p_e^π is a proposal's nominal successful-category probability averaged under $\mu_{\mathcal{K}}$, and

$$q_e(z) = \frac{b_e(z)_{\alpha_e}}{L_*} \tag{6.3}$$

for each base channel, then

$$\sum_{e \text{ base}} |p_e^\pi - q_e(z)| \le B_*\zeta_{2N}(\rho). \tag{6.4}$$

Here the point-particle uniform averages are exactly q for both contact binding and release success, so (3.2) cancels this factor. A zero-radius unary placement succeeds. To verify the finite-radius error, compare each average before conditioning with its zero-radius version, then condition on current feasibility. The difference is supported on a wall layer or a pair-exclusion ball in the joint old/new slots. Integrating the other coordinates gives the three-dimensional volume bounds just stated. Sum the coefficient amplitudes; their sum is at most L_* before the spare factor 2^{10} in B_*. This proves (6.4) without an invariant-volume computation.

All categories other than successful base channels will be placed in a single reference residual category 0. In particular no claim about the stationary intensity of rejection marks is needed. The reference probability of 0 is $1 - \sum_e q_e(z) \ge 0$.

6.2 Why proposals remain independent of Brownian innovations

Fix any admitted realization. By adding dummy proposals using fresh independent randomization, augment its actual attempts to a counting process N^* with deterministic intensity $\lambda = L_*/h$. This augmentation does not alter any physical event or the policy's input history. Each actual attempt remains an event of N^*; the added events are proof-only dummies.

For completeness, N^* has compensator $\lambda\,ds$, its jumps have size one, and its continuous cross-variation with the Brownian innovation is zero. The exponential martingale for (W, N^*) gives, for

every time increment u,

$$\mathbb{E}[e^{i\theta\cdot(W_{s+u}-W_s)+iv(N^*_{s+u}-N^*_s)} \mid \mathcal{F}_s] = e^{-u|\theta|^2/2+\lambda u(e^{iv}-1)}. \tag{6.5}$$

One obtains the identity first after localization by the jump–Itô formula and then removes localization, since the intensities are bounded. Thus the joint innovations are independent of the full past and their two components are independent. The next proposal gap is exponential of rate λ and is independent of future Brownian innovations. Reflection is finite variation and does not change this calculation. Actual physical attempts themselves were never assumed to form a constant-rate process.

6.3 Couple categories, preserving the actual hidden history

Suppose reference and actual fine counts agree just after a proposal. Condition on the preceding complete history $\mathcal{H}$ and the next gap g. Let E be the actual successful base-channel category, with every steric failure or dummy assigned category 0. Denote its conditional law by $p(\mathcal{H}, g)$. Apply proposition 5.3 to the normalized channel predicates. After summing amplitudes, (6.4) and the chemical multiplier error give

$$d_{\mathrm{TV}}(p(\mathcal{H}, g), q(z)) \le b(g) + W\delta^{2/9} + \delta + B_*\zeta_{2N}(\rho). \tag{6.6}$$

The residual coordinate costs no additional factor in total variation: its absolute discrepancy is at most the sum over successful categories.

Take a maximal coupling $\pi(e, f)$ of these category laws. After the actual segment and category $E = e$ have occurred, draw the auxiliary reference category F with conditional probabilities $\pi(e, f)/p_e$; zero-mass rows are irrelevant. Use an independent uniform random variable. Leave the actual path, selected tuple, offset, failure type, contact clocks and all future policy decisions untouched. Summing a row gives the true actual law; summing a column gives $q(z)$ conditional on $\mathcal{H}, g$. This is a disintegration of a coupling, not a physical resampling operation.

On equal categories the fine counts and the exact proposal times agree. Reference nonzero categories are the base kinetic channels, including private channels and genuine nulls. Reference category zero leaves the fine count unchanged. Consequently, after time is divided by h, the reference count-and-base-mark law is exactly the nominal kinetic law with coefficients b_e. After the first category mismatch it can be continued according to that same law. Its marginal remains correct. In particular the argument does not try to couple a fault policy evaluated on invented equilibrium positions.

6.4 Pay every initial and post-event layer

Let $S = hT$ be the physical horizon and let G_j be successive proposal gaps. A renewal bound gives

$$\mathbb{E}\sum_{T_j\le S} b(G_j) \le \lambda I(1+\lambda S) = \frac{L_*I}{h}J. \tag{6.7}$$

To check it without a stopping-index independence assumption, sum over the preceding proposal time u. The next gap density is $\lambda e^{-\lambda g}$, and integrating $b(g)$ over $0 < g \le S - u$ is at most λI. There is one initial preceding time and expected λS later preceding times. This proves (6.7). Also $\mathbb{E}N^*(S) = L_*T \le J$.

The union bound on the first mismatch yields

$$E_{\mathrm{spatial}} \le \frac{L_*I}{h}J + JW\delta^{2/9} + J\delta + JB_*\zeta_{2N}(\rho). \tag{6.8}$$

The first gap starts at the specified initial placement. Every later gap starts at its actual post-event placement. The terminal gap is also part of the coupled record: if it contains no further proposal there can be no unmatched physical mark in it. There is no deletion of that gap, or of any residence time. Observing the decoder and hiding private marks contracts total variation. Adding the kinetic error $\epsilon/8$ will prove (2.12).

7 Compiler choices and all-run resources

We now give the algorithm in its dependency order. Compute the paired kinetic output of proposition 3.1, append the structural rejection list, compute V, q by fixed-dimensional rational geometry, and form T, L_*, J, B_* in (6.1). Choose

$$a = \min\Big\{2^{-30}(2+N+F+\kappa^{-1})^{-20}, \frac{\min(\kappa, r)}{2^{16}(1+N+P_{\max})}, \\ \frac{\epsilon\kappa^3}{2^{20}JB_*(1+N)^4}, \frac{1}{2}\Big\}, \qquad \rho = a^2. \tag{7.1}$$

All these numbers are rational. This choice satisfies the radius ceiling and (4.4). Moreover

$$\zeta_{2N}(\rho) \le 32\rho(1+N)^4\kappa^{-3}, \qquad JB_*\zeta_{2N}(\rho) \le \epsilon/64.$$

Evaluate (5.1)–(5.2) at this radius, and choose

$$h = \left\lceil \max\{1, q^{-1}, 64L_*IJ/\epsilon\} \right\rceil, \qquad H = hH_{\mathrm{kin}}, \tag{7.2}$$

$$\delta = \min\left\{\tfrac{1}{4}, \tfrac{1}{2}\delta_{\mathrm{kin}}, \left[\frac{\epsilon}{64J(1+W)}\right]^9\right\}. \tag{7.3}$$

Emit (3.2) and its inherited rejection coefficients. They lie in $(0, 1]$. Also $h \ge L_*$, so actual total intensity is at most one. Every perturbed positive coefficient has inverse at most $(4/3)c_e^{-1}$. An instantaneous spatial propensity may of course vanish when its geometric predicate is false; no reciprocal bound for such a propensity is asserted.

Substituting these choices into (6.8) gives a mixing error at most $\epsilon/64$, a stationary error at most $\epsilon/64$, and

$$JW\delta^{2/9} \le \epsilon/64, \qquad J\delta \le \epsilon/64.$$

Together with $\epsilon/8$ from proposition 3.1 this is strictly below ϵ. The final observation map divides physical times by hH_{kin}, retaining every original mark. This proves the accuracy part of theorem 2.1. The construction has no requirement involving ρ times the enlarged horizon hT.

7.1 What the physical resource charges mean

Let $F_{\mathrm{all}}(s)$ count every actual attempt by physical time s, including success, failure, hidden, original and global null events. Sum continuous particle quantities over their actual lifetimes. Define unsigned push lengths

$$W_{\mathrm{wall}}(s) = \sum_{i,f}\int_0^s |D_{s(i)}(u, Q_i)n_f|\, dL_{if}(u),$$

$$W_{\mathrm{pair}}(s) = \sum_{i<j}\int_0^s (|D_{s(i)}e_{ij}| + |D_{s(j)}e_{ij}|)\, dL_{ij}(u). \tag{7.4}$$

These definitions sum simultaneous contacts separately and cover every admitted decomposition. They are lengths, rather than thermodynamic energies or finite counts of Brownian contacts.

Theorem 7.1 (Actual all-run resources)**.** *For every admitted realization, at every physical time $s \geq 0$,*

$$|Z_s|_1 \leq N, \qquad \mathbb{E}F_{\mathrm{all}}(s) \leq (L_*/h)s, \qquad \mathbb{E}F_{\mathrm{all}}(H\tau) \leq L_*T, \tag{7.5}$$

$$\int_0^s |Z_u|_1 \, du \leq Ns, \qquad \sum [Q]_s \leq 9Ns, \tag{7.6}$$

$$\sum \int_0^s |v_i| \, du \leq \delta N s, \qquad \sum \int_0^s |g_{s(i)}| \, du \leq 3\Lambda N s. \tag{7.7}$$

The total reaction-placement displacement is at most $\sqrt{3}rF_{\mathrm{all}}(s)$; total radius variation, including birth and removal, is at most $4\rho P_{\max}F_{\mathrm{all}}(s)$. For a universal constant $C_0 \leq 2^{80}$,

$$\mathbb{E}(W_{\mathrm{wall}}(s) + W_{\mathrm{pair}}(s)) \leq C_0 \Big[\rho(1 + \mathbb{E}F_{\mathrm{all}}(s)) + \frac{N(1+N)^4(1+\Lambda)^3}{\kappa^2 \rho d_0^7} \, s \Big]. \tag{7.8}$$

The process is nonexplosive. At $s = H\tau$ all these expected resource charges, and all numerical/compiler charges in theorem 2.1, satisfy (2.13). One sufficient coarse choice is $p = 2^{32}$ and $C = 2^{2^{40}}$.

Proof. Every successful reaction conserves $\omega \cdot PZ = N$, and every failed attempt preserves Z. Since $\omega_i \geq 1$ and every payload is nonzero, $|Z|_1 \leq \sum_s d_s Z_s \leq N$. Bounded all-attempt intensity gives (7.5) and nonexplosion, without conditioning on accurate runs. Equation (7.6) follows from $2 \operatorname{tr} A \leq 2(3n)(1 + \delta d_0) \leq 9n$. The drift bounds follow from their definitions. A release or binding uses only an offset in $[-r, r]^3$; retained-slot changes have zero displacement. There are at most two slots on each side, proving the placement and radius charges.

For (7.8), apply the core potential of lemma A.1 across successive particle lifetimes. Each actual contact lowers it by at least half the core-tightening parameter per clock increment. Its jump at a chemical attempt is bounded by its full range. The resulting estimate (A.9) controls all wall and pair clocks; each contributes push length at most one or two times its clock increment. Substituting $\eta = \eta_{\max}$ and $\alpha^{-2} = 2^{20}(1+N)^4/\kappa^2$ gives (7.8). The localization is removed by monotone convergence of stopped clocks; thus this bound cannot be evaded by a different simultaneous-contact decomposition. The explicit polynomial and finite-bit accounting is in appendix C. □

The smaller radius in (7.1) increases the permissible work bound and short-time mixing costs. These costs remain polynomial in the numerical charge; they need not be small. In particular, a vanishing excluded-volume fraction is not being used to claim vanishing total collision influence.

7.2 Chosen data and proof objects

The emitted controls are the reaction structure, payload, activities, coefficients, radius, slowdown and tolerance. The vessel and transport bounds are supplied data. The fields D_s are supplied only by the physical environment, within the contract. The uniform measure, $\mathcal{K}$-volume, core minimizer, buffer flow, ranks, intermediate paths, nominal transition functions, augmented proposals and maximal coupling are analytical objects. None is sampled, optimized or solved by the compiler or imposed as a physical steering mechanism.

8 Scope, obstructions and further questions

The theorem converts a chemical trajectory specification into a local finite-radius model even when the detailed transport profile is unavailable. It separates nominal equilibrium consistency from finite-time trajectory accuracy and controls the latter from arbitrary feasible placements. Its structural graph assertion is exact, while its timed-path assertion is a total-variation approximation on a specified horizon.

A cage invalidates a tempting simpler proof. Small radii do not imply that a particle mixes rapidly with all others frozen. In local coordinates at a cubic corner, let two particles have equal radius a, let the first center range over the positive orthant, and put the other center at (h, h, h) with $h = 5a/4$. The corner is feasible because $\sqrt{3}h > 2a$. Yet every point of the separating simplex $u_1 + u_2 + u_3 = h$, $u_i \geq 0$, lies in the exclusion ball: its farthest vertices have distance $\sqrt{2}h < 2a$ from (h, h, h). The first particle is trapped in a corner component when the other is fixed, at arbitrarily small a. This does not trap the all-mobile dilute configuration space. It explains why a conditional one-particle Poincaré argument is invalid and why the joint buffer/separation construction is useful.

Boundary of the compiler. The physical dimension is fixed at three, while chemical dimension and expanded reaction order vary. The input must have the supplied positive integer conservation law and detailed-balance certificate. A general open-network spatial realization does not follow from the open kinetic theorem. Nor does the proof cover a fixed positive packing fraction, macroscopic radii independent of the task, a different reflection law, unbounded time horizons at the same accuracy, or interventions outside the specified fault box.

The numerical charge contains high-order propensities and activity reciprocals. The resulting reciprocal radius, clock and precision may be severe. A polynomial in this charge is not a polynomial in binary input length alone or an affordability guarantee. The construction specifies ideal channels; selecting actual molecules, suppressing off-graph leakage, realizing the transport and contact laws, and accounting for chemical free energy remain separate problems. No molecular synthesis, measured trajectory or practical resource advantage is inferred.

Constructive questions. One direction is a quantitative geometry theorem beyond the dilute regime, where connectivity and metastable bottlenecks become genuine issues. Another is to replace hard exclusion by a physically specified repulsion with a comparable conditional transport theorem. Extending the compiler to open networks would require a paid capacity and geometry argument compatible with particle creation, not a formal invocation of the conservative proof. Material realization would additionally need rates and leakage estimates for chosen reaction mechanisms. These questions concern new operations; they are not corollaries of the present path approximation.

Acknowledgements and AI use

The mathematical ideas, constructions, proofs, and manuscript were developed by GPT-6 Astra, with additional AI assistance for review. Human input was minimal and consisted primarily of setting the research objectives and providing occasional feedback.

A Reflected motion: estimates, construction and smoothing

This appendix proves lemma 5.1 and the contact-work estimate. All norms on coefficient derivatives are operator norms; summing the trace of a joint covariance costs a factor $3n$, including when its blocks are correlated. The constants below are deliberately oversized fixed bounds.

A.1 A tightened-core potential

Let α be (4.2), set $M_0 = A_0^{-1}$, and put

$$\eta_{\max} = 2^{-20}\frac{\alpha^2\rho d_0^3}{1+\Lambda}. \tag{A.1}$$

Unlike the buffer parameter in (4.8), this parameter is chosen afresh to estimate a given time interval.

Lemma A.1 (Core potential)**.** *For $0 < \eta \le \eta_{\max}$, define*

$$\Phi_\eta(t,q) = \frac{1}{2}\min_{y\in\mathcal{K}_\eta}(q-y)^\top M_0(t,q)(q-y). \tag{A.2}$$

The minimizer is unique on a neighborhood of $\mathcal{K}$, the potential is $C^{1,1}$ in space and Lipschitz in time, and

$$0 \le \Phi_\eta \le \frac{2\eta^2}{\alpha^2 d_0}, \qquad |q-y_*| \le \frac{2\eta}{\alpha\sqrt{d_0}}, \tag{A.3}$$

$$|\nabla\Phi_\eta| \le \frac{8\eta}{\alpha d_0^{3/2}}, \qquad \left\|\nabla^2\Phi_\eta\right\|_{\mathrm{op}} \le 2^{16}(1+\Lambda)^2 d_0^{-4}, \tag{A.4}$$

$$|\partial_t\Phi_\eta| \le \frac{8\Lambda\eta^2}{\alpha^2 d_0^3}. \tag{A.5}$$

The Hessian and time derivative are interpreted almost everywhere. For each individual active constraint,

$$\nabla\Phi_\eta \cdot A_0\nabla g_j \le -\eta/2. \tag{A.6}$$

No minimizer in (A.2) is evaluated by the compiler.

Proof. Move $2\eta/\alpha$ in the common inward direction of lemma 4.1. For near pair gaps use convexity of the norm; wall gaps are affine. Far gaps decrease by at most $\sqrt{2}$ times the displacement. Equation (A.1) therefore implies $\operatorname{dist}(q,\mathcal{K}_\eta) \le 2\eta/\alpha$, proving nonemptiness and (A.3).

At a minimizer let $z = q - y_*$. The cone is a constraint qualification and gives multipliers

$$M_0 z = -\sum_j \lambda_j \nabla g_j(y_*), \qquad \lambda_j \ge 0, \qquad \sum_j \lambda_j \le |M_0 z|/\alpha. \tag{A.7}$$

For an active wall and $y' \in \mathcal{K}_\eta$, its gradient paired with $y' - y_*$ is nonnegative. For an active pair, expansion of the squared pair distance gives

$$\nabla g_j(y_*)\cdot(y'-y_*) \ge -|y'-y_*|^2/(2\rho). \tag{A.8}$$

Consequently the increase in the quadratic objective is at least

$$\left(\frac{1}{2} - \frac{|z|}{2\alpha\rho d_0}\right)|y'-y_*|^2.$$

It is strictly positive by (A.1), proving uniqueness even though the core is nonconvex.

Apply (A.8) at the minimizers for two query points, add the inequalities, and expand their two inverse metrics. Absorbing the quadratic remainder gives

$$|y_*(t,q) - y_*(t,q')| \le 4d_0^{-1}|q-q'|.$$

Here $\|M_0(q) - M_0(q')\|_{\mathrm{op}} \le 8\Lambda d_0^{-2}|q-q'|$; its product with the displacement in (A.3) is absorbed using (A.1). This argument also applies in a small open neighborhood. The envelope derivative is

$$\nabla\Phi_\eta = M_0 z + e, \qquad e_h = \tfrac{1}{2} z^\top (\partial_h M_0) z.$$

The inverse differentiation identities

$$\partial M_0 = -M_0(\partial A_0)M_0, \qquad \partial^2 M_0 = M_0(\partial A_0)M_0(\partial A_0)M_0 + M_0(\partial A_0)M_0(\partial A_0)M_0 - M_0(\partial^2 A_0)M_0$$

(with the two derivative orders in the first two terms) and the preceding Lipschitz estimate give (A.4). These bounds use the block structure of A_0 and fixed three-dimensional block derivatives; no coordinatewise dimension-exponential norm is used. The time envelope derivative is $z^\top(\partial_t M_0)z/2$, giving (A.5).

At a pair contact, put $u = q_i - q_j$ and let u' be its core counterpart. Since $|u'| \ge |u| + \eta$, expansion of squares yields

$$\nabla g_j(q)\cdot(y_* - q) \ge \eta - |y_* - q|^2/(2\rho).$$

For walls the quadratic term is absent. The main envelope term paired with $A_0\nabla g_j$ is therefore at most $-\eta + |z|^2/(2\rho)$. The remaining term is at most $2^6\Lambda d_0^{-2}|z|^2$. Using (A.1) gives (A.6). □

Corollary A.2 (Clock bounds)**.** *For every admitted realization, summing all lifetime clocks,*

$$\mathbb{E}\mathcal{L}(s) \le 2^{24}\left[\frac{\eta}{\alpha^2 d_0}(1 + \mathbb{E}F_{\mathrm{all}}(s)) + \frac{N(1+\Lambda)^2}{\eta d_0^4}s\right]. \tag{A.9}$$

On a chemical-free interval of length $u \le \eta_{\max}^2$,

$$\mathbb{E}[\Delta\mathcal{L} \mid \mathcal{F}_s] \le P_{\mathrm{loc}}\sqrt{u}, \qquad P_{\mathrm{loc}} = 2^{24}\{\alpha^{-2}d_0^{-1} + N(1+\Lambda)^2 d_0^{-4}\}. \tag{A.10}$$

Proof. Apply Itô's formula to (A.2). Its continuous bulk term is bounded by $2^{20}N(1+\Lambda)^2 d_0^{-4}$: the covariance term is $A : \nabla^2\Phi_\eta$, bounded by $4n\,\|\nabla^2\Phi_\eta\|_{\mathrm{op}}$, and the drift and time terms fit the same bound. Every contact contributes at most $-\eta/2$ times its clock increment. Each chemical event changes the potential by at most its full range in (A.3), uniformly over old and new particle lists. Stop at bounded total clock and event count, take expectations, and use $\Phi_\eta \ge 0$. This gives (A.9). Monotone convergence removes the stopping times. Choosing $\eta = \sqrt{u}$ proves (A.10) conditionally.

The $C^{1,1}$ formula can be justified by spatial and time mollification on the neighborhood in lemma A.1. At contact the gradient inequality persists in the limit; Hessians stay bounded. The process spends zero Lebesgue time on any constraint surface, by the one-dimensional occupation formula for its gap and uniform ellipticity. Thus the weak Hessian version gives the same Itô identity. This also justifies using time-Lipschitz profiles. No regularity of the fault policy was invoked. □

Conditional iteration of (A.10) gives $\mathbb{E}[(\Delta\mathcal{L})^k \mid \mathcal{F}_s] \le k!(P_{\mathrm{loc}}\sqrt{u})^k$. For instance, expand the power as an ordered Stieltjes integral and condition successively on its preceding times. In the notation (5.1), $P_{\mathrm{loc}} \le 2^{48}Z_*^8 < \mathsf{M}$ and $\eta_{\max}^{-1} \le 2^{42}Z_*^{11}$. Therefore $\ell \le \eta_{\max}^2$ and $8\mathsf{M}P_{\mathrm{loc}}\sqrt{\ell} \le 8\mathsf{M}^{-6} < 1/2$. Summing the exponential series proves lemma 5.1(i), also at stopping times.

A.2 Short conditional comparison

Couple nominal and actual motion from the same point, using the same $3n$-dimensional Brownian innovation, and set $z = q - q'$. Test their difference with $E = z^\top M_0(t,q)z$. For a pair contacting in one path, feasibility of the other gives (A.8); wall terms have the favorable sign apart from coefficient differences. The first and second spatial derivatives of M_0 control those differences. The intrinsic drifts are Lipschitz. For the square roots, the Sylvester equation

$$A^{1/2}(A^{1/2} - A_0^{1/2}) + (A^{1/2} - A_0^{1/2})A_0^{1/2} = A - A_0$$

gives $\left\|A^{1/2} - A_0^{1/2}\right\|_{\mathrm{op}} \le 2\delta\sqrt{d_0}$. Summing its Frobenius square costs $3n$. Itô differentiation consequently gives, after localization,

$$dE \le KE(\,ds + d\mathcal{L} + d\mathcal{L}') + Kn\delta^2\,ds + dM, \tag{A.11}$$

$$d[M] \le K(E^2 + n\delta^2 E)\,ds, \qquad K \le 2^{80}Z_*^{80} < \mathsf{M}. \tag{A.12}$$

Terms linear in $\delta\sqrt{E}$ were absorbed by $E + n\delta^2$. In particular no covariance-fault derivative occurs; the nominal reflection is essential for this bound.

Here is the moment step explicitly. For $n > 0$, $\delta > 0$, set $a_* = n\delta^2$ and $Y = E + a_*$. Equations (A.11)–(A.12) imply $dY \le 2KY(\,ds + d\mathcal{L} + d\mathcal{L}') + dM$ and $d[M] \le KY^2\,ds$. Thus $R = \int Y^{-1}\,dM$ has $[R]_u \le Ku$. The logarithmic Itô formula gives

$$Y_{s+u} \le a_* \exp\{2K(u + \Delta\mathcal{L} + \Delta\mathcal{L}') + R_u - \tfrac{1}{2}[R]_u\}.$$

For its second moment, Cauchy–Schwarz separates the clock exponential from $e^{4R-2[R]}$. The latter has expectation at most e^{6Ku}, since $e^{4R-8[R]}$ is a localized exponential martingale and $[R] \le Ku$. The former is bounded by the conditional clock estimate; for the sum of two clocks use Cauchy–Schwarz once more. All required parameters are below $8\mathsf{M}$ after enlarging the raw polynomial constant within M, and $u \le \ell$. Removing localization gives

$$\mathbb{E}[E^2 \mid \mathcal{F}_s] \le \mathsf{M}^6\delta^4, \qquad \mathbb{E}[|q - q'| \mid \mathcal{F}_s] \le \mathsf{M}^4\delta.$$

This proves lemma 5.1(ii). At $\delta = 0$, use $Y = E + a_*$ with arbitrary $a_* > 0$ and let $a_* \downarrow 0$ in the same bound. This gives equality of positions for two nominal solutions driven by the same Brownian motion. It does not require equality of individual clock decompositions.

A.3 Existence at corners and identification of the form

We describe the approximation, since a theorem for normal reflection alone would not establish (2.8). Extend each D_s to a neighborhood of the eroded vessel by convolution at width smaller than $\rho/4$, and mollify in time. This uses values strictly inside Ω, preserves ellipticity and the stated derivative bounds, and approximates the original fields locally uniformly with the needed derivatives.

Smooth the intersection of constraints with

$$g_\varepsilon = -\varepsilon \log \sum_j e^{-g_j/\varepsilon}, \qquad \mathcal{K}^\varepsilon = \{g_\varepsilon \ge 0\}.$$

Pair gaps are smooth in a fixed neighborhood of $\mathcal{K}$; a harmless smooth extension away from this neighborhood defines the displayed expression globally. Since $\min_j g_j - \varepsilon \log J_c \le g_\varepsilon \le \min_j g_j$, these

are inner approximants exhausting the interior. At a boundary point, constraints with appreciable soft weight have gaps at most $O(\varepsilon \log J_c)$ and share the contact cone. For small ε,

$$|\nabla g_\varepsilon| \geq \alpha/2, \qquad \nabla^2 g_\varepsilon = \sum_j w_j \nabla^2 g_j - \varepsilon^{-1} \operatorname{Cov}_w(\nabla g_j) \leq (4/\rho) I. \tag{A.13}$$

Thus the negative part of the outward second fundamental form is bounded by $8/(\alpha\rho)$, independently of corner rounding. Large positive curvature at a rounded corner is harmless.

On a smooth approximant with smooth coefficients the conormal equation can be constructed by an inward penalty supported in a shrinking boundary band. Its coefficients, for a fixed band width, are bounded and Lipschitz, so Picard iteration constructs its unconstrained equation. Itô's formula for squared exterior distance makes time outside the band vanish in the limit. The core potential gives the uniform bounds on the integrated penalty in that band, where (A.6) holds with, say, $-\eta/4$. Hence subsequential limits are constrained and their finite variation term is the stated inward conormal. This constructs the smooth reflected process without using a configuration-space algorithm.

Decompose its boundary normal using the soft weights in (A.13). Absorb the scalar normal normalization into nondecreasing clocks for each constraint. The clock moment argument is uniform for sufficiently small ε; constraints carrying nonnegligible weight have gap tending to zero. Its higher conditional moments give tightness of positions and clocks on compact time intervals: $\mathbb{E}[(\Delta\mathcal{L})^k] \leq k! C^k u^{k/2}$ for arbitrary fixed k, and Brownian increments have the same power. Approximating initial points may therefore tend to *any* point in $\mathcal{K}$, including simultaneous contacts. In a limit the clocks have their proper supports and the conormal coefficients converge. The limiting equation is (2.8) at zero faults. The comparison with $\delta = 0$ proves position uniqueness and identifies the whole limit, rather than just a chosen subsequence. Redundant constraints can leave clocks nonunique; all their unsigned costs still obey (A.9).

We also identify the invariant measure, rather than presuming it from the SDE notation. On each smooth approximant integration by parts gives the Neumann form with Lebesgue reference measure. These forms converge to (3.3). To verify the two variational directions, restrict a smooth extension of an $H^1(\mathcal{K})$ function to the inner domains for the recovery direction. Such extensions are dense because of the local cone. For the lower-bound direction, uniform cone charts provide extensions from the approximants to a common neighborhood with bounded H^1 norm for each fixed problem, uniformly in ε. Weak compactness and the interior exhaustion give the lower bound on the limiting energy. Alternatively these two directions follow by applying the small inward flow used in theorem 4.2 at a time tending to zero on each chart. No quantitative chart-count constant enters a final estimate.

The weak heat equation is unique: the difference of two solutions has nonincreasing squared L^2 norm by ellipticity. Time-step products of the frozen forms, followed by time refinement, give the same weak solution; the common form domain is $H^1(\mathcal{K})$ and the time-Lipschitz coefficient bounds control the refinement error in its energy norm. The preceding path limits identify this solution with the nominal transition evolution. Every frozen step preserves uniform measure, hence so does the limit. For two different initial points the same comparison calculation, with $E(0)$ retained, bounds the short-time mean distance by a constant times their initial distance. It gives a Feller extension to every boundary point. The form process and the constructed process agree for Lebesgue-almost every starting point by the weak equation, hence everywhere by this continuity. This proves the existence and form claim in lemma 5.1 and proposition 3.2.

Finally interlace these fixed-count motions with the bounded-rate local attempt kernels. Every successful placement is feasible by definition, so the next motion starts in the just-constructed

all-boundary family. The proposal bound excludes accumulation of jumps. This constructs the full nominal particle process, not just its separate fixed-count pieces.

A.4 Local smoothing with controlled constants

On a smooth approximant give configuration space the metric $g_t = A_0(t,q)^{-1}$. Direct differentiation of the metric volume gives

$$\operatorname{div}(A_0\nabla) = \Delta_{g_t} + \mathcal{Z}_t \cdot \nabla, \qquad \mathcal{Z}_t = \tfrac{1}{2} A_0 \nabla \log\det A_0. \tag{A.14}$$

Its unit inward metric normal is the normalized nominal conormal, with the normalization absorbed by the local time. Inverse differentiation, (A.13) and the supplied third spatial derivatives give

$$\operatorname{Ric}_{g_t} - \nabla^{g_t}\mathcal{Z}_t - \tfrac{1}{2}\partial_t g_t \geq -K_g g_t, \qquad \mathrm{II}_{g_t} \geq -K_g g_t, \qquad K_g \leq 2^{80} Z_*^{80}. \tag{A.15}$$

To track dimensions, each metric block depends only on its own three coordinates. Products of inverse metrics and their first two derivatives have degree at most eight in d_0^{-1} and $1+\Lambda$; the drift derivative uses at most the third derivative of D_s. The curvature contractions cost at most $(3n)^2$. The boundary bound adds $\alpha^{-1}\rho^{-1}$ and first metric derivatives. All are beneath the displayed coarse bound. The metric local time is at most a fixed polynomial multiple of the sum of constraint clocks, also beneath that bound. These estimates are uniform in the smoothing parameters.

Apply the derivative formula of [3, Theorem 2.5] with the linear control function on an interval of length u. Its multiplicative functional has norm bounded by the exponential of the curvature time integral plus the negative-boundary-curvature local-time integral. The exponential clock estimates above bound its second moment by a universal constant on $u \leq \ell$. Cauchy–Schwarz in the formula gives

$$|\nabla P^0_{s,s+u} f| \leq \frac{\mathsf{M}^2}{\sqrt{u}} \|f\|_\infty .$$

For $u \in [\ell/2, \ell]$ this is at most $\mathsf{M}^{11}\|f\|_\infty$. Local cone paths convert the intrinsic bound to a local Euclidean Lipschitz bound with an additional factor at most M. The local comparison radius is at least $u_0 = \mathsf{M}^{-1}$, by the same cone charts as in (5.5). For points farther apart than u_0, boundedness gives a Lipschitz ratio at most $2\mathsf{M}\|f\|_\infty$, also below $G\|f\|_\infty$. Thus G in (5.2) is a global Euclidean bound. The argument works for a centered smoothed future function as well: both its local gradient and its far-point ratio acquire the same decaying supremum-norm factor. This is the bound needed in conditional telescoping.

On a smooth compact approximant the bounded-gradient hypothesis in that formula holds first for smooth data satisfying the Neumann condition. Approximate bounded data after a positive terminal smoothing interval and pass to the limit; the estimate depends only on the supremum norm, not on this auxiliary smoothing. Uniform equicontinuity, weak convergence of the paths and bounded continuous approximation of measures then pass the estimate to $\mathcal{K}$ and bounded measurable terminal data. Time mollification handles the original Lipschitz time dependence. This proves lemma 5.1(iii) with every boundary and time hypothesis paid.

B Observable tubes and conditional telescoping

B.1 Three-dimensional tube bounds

Let $\nu_0 = \bigotimes_i \mathrm{Unif}(\Omega)$. Convexity and the inball imply

$$\frac{\mathrm{Vol}\{x \in \Omega : \operatorname{dist}(x, \partial\Omega) < s\}}{V} \leq 3s/\kappa \quad (0 \leq s \leq \kappa).$$

Indeed the contraction $(1 - s/\kappa)\Omega + (s/\kappa)c$ lies at least s inside every supporting plane. For any pair of uniform centers, conditioning on one bounds exclusion probability by $4\pi(r_i + r_j)^3/(3V)$. Since $V \geq \kappa^3$, $\sum_i r_i \leq \rho N$ and $\sum_{i<j}(r_i + r_j)^3 \leq \rho^3 N^4$, a union bound gives

$$\nu_0(\mathcal{K}^c) \leq 3\rho N/\kappa + 5\rho^3 N^4/\kappa^3 < 1/2.$$

Thus $\mu_{\mathcal{K}} \leq 2\nu_0$. The same conclusion for the product of eroded vessels used in theorem 4.2 follows by omitting the wall term and using their volumes at least $V/8$; (4.4) absorbs this factor.

A slab of width $2w$ has volume at most $32w$ in the containing cube, uniformly in its orientation. A shell $\{x : ||x - y| - R| \leq 2w\}$ with $R \leq 2\rho N$ and $w \leq 1/2$ has volume at most $2^{10}w$. Divide by $V \geq \kappa^3$ when the integrated coordinate is a current center. When it is a release offset, divide by v. These bounds also cover an affine shifted center $x + z$ by conditioning on the other coordinate. Cube-contact faces are slabs. Conditional density of the remaining coordinates is not invoked: domination by $2\nu_0$ is applied to the full joint integral before integrating unused coordinates. At most $2F + 4N + 6$ surfaces affect any elementary channel. Replacing all thresholds by width-w ramps therefore gives the explicit safe bounds

$$\mathrm{Lip}(f_w^{\pm}) \leq 2^{12}(1 + N + F)/w, \qquad \mu_{\mathcal{K}}(f_w^+ - f_w^-) \leq 2^{20}(1 + N + F)\kappa^{-3}(1 + v^{-1})w.$$

They are beneath (5.6). Non-strict versus strict feasibility on the threshold surfaces is covered by the two ramps, including boundary starting points.

For stationary rate bias one uses full wall layers, not a sum of F unrelated facet layers. This is why (6.2) has no facet-count factor. Include the joint old and proposed slots, whose total payload is at most $2N$. Pair-ball volume, wall contraction, offset density and contact amplitudes give a fixed multiple of ζ_{2N}, paid by B_* in (6.4). Thin individual facets and redundant inequalities do not introduce inverse-angle or inverse-area costs.

B.2 The telescoping identity with a non-Markov actual motion

For a grid $t_0 < \cdots < t_k$, a bounded terminal function f and $f_j = P^0_{t_j,t_k} f$, the exact identity is

$$\mathbb{E} f(Q_{t_k}) - P^0_{t_0,t_k} f(Q_{t_0}) = \sum_{j=1}^{k} \mathbb{E}\left[f_j(Q_{t_j}) - P^0_{t_{j-1},t_j} f_j(Q_{t_{j-1}}) \right]. \tag{B.1}$$

It follows by cancellation of adjacent expectations, using the nominal composition law only. For each summand condition on the entire actual history at t_{j-1} and use the short coupled estimate; the nominal comparison starts at that actual point. This is why no Markov property of the actual process is required.

For a ramp $f_w^{\pm}$, the final summand is bounded by $\mathsf{M}^5\delta/w$. For an earlier summand, split off an initial nominal smoothing interval of length $\ell/2$. Constants have zero gradient, so lemma 5.2 and lemma 5.1(iii) bound its gradient by $Gb((t_k - t_j - \ell/2)_+)$. Gaps of the grid are at least $\ell/2$. Summing the decreasing envelope is at most $4 + 4I/\ell$. This proves the estimate in proposition 5.3 without a term proportional to the full horizon.

If the gap length is the next independent exponential proposal gap, (6.5) permits disintegration at each value of that length. The conditioned Brownian innovations remain Brownian relative to the past needed by the predictable coefficients. The conditional clock, comparison and telescoping estimates are therefore the same. To use a fixed innovation dimension throughout births and removals, keep $3N$ Brownian coordinates and assign unused slots predictably; fresh increments provide each new lifetime's innovation. The unused coordinates do not affect the positions. This also makes (6.5) literal across changes of particle count.

C Finite-bit geometry and polynomial majorants

C.1 Exact fixed-dimensional volume

The compiler computes only V in dimension three and

$$\mathrm{Vol}\{(x,y)\in\Omega^2 : -r\le y_j-x_j\le r,\quad j=1,2,3\}$$

in dimension six. All defining inequalities in this calculation are rational; unit facet normals are unnecessary. In a fixed dimension k, enumerate k-subsets of boundary hyperplanes, solve each nonsingular rational linear system and retain feasible intersection points. Deduplicate points and faces using exact rational comparisons. Enumerate faces by hyperplane incidences and deduplicate their vertex sets. A barycenter of the vertices of a face lies in its relative interior. Flags of faces, with these rational barycenters, triangulate the boundary and then the polytope. Sum absolute simplex determinants divided by $k!$.

This handles redundant inequalities, nonsimple vertices and degeneracies without a general-position assumption. For fixed $k\le 6$, the numbers of candidate vertices, faces and flags are fixed powers of the facet count; Gaussian elimination and determinant arithmetic have polynomial bit cost. Lower-dimensional faces contribute zero full-dimensional volume. The input promises ensure both requested full-dimensional volumes are positive. The volume quotient is q in (3.1). No operation has dimension $3N$. Normalized facet directions in the physical law can be represented by their rational coefficients and the positive square root of their rational squared norm; this is finite algebraic syntax, not an exact-real oracle.

C.2 Why the local ledger has fixed powers

For clarity, the intermediate bounds used to dominate the analytic estimates by (5.1) are listed here. They follow from the explicit derivatives and inequalities above; all numerical constants are independent of population and facet arrangement.

Quantity	Sufficient bound
α^{-1}, number of constraints	$2^{10}Z_*^3,\ 2Z_*^2$
$\eta_{\max}^{-1}$	$2^{42}Z_*^{11}$
Short clock coefficient P_{loc}	$2^{48}Z_*^{8}$
Inverse-metric derivatives through order two	$2^{16}Z_*^{6}$
Comparison coefficients before moment estimates	$2^{80}Z_*^{80}$
Curvature and metric-local-time conversion, separately	$2^{80}Z_*^{80}$
Finite products needed in the local moment/smoothing proof	$2^{512}Z_*^{512}$
Tube coefficient and ramp numerator	$2^{30}Z_*^{12}$

To verify the last product bound, the comparison moment step uses at most four powers of the comparison coefficient, two inverse-ellipticity factors and the trace factor. The curvature exponent uses the product of its bound and the metric-clock conversion. Both have degree below 512 and constants below 2^{512}. The only time derivatives used are the first derivatives of the mollified metric, uniformly bounded by the supplied Lipschitz constants. No high derivative of a rounded boundary is charged: the derivative formula uses its second fundamental form's lower bound, which is (A.13). Increasing positive curvature does not worsen it. The master M strictly dominates this entire finite ledger. The powers in (5.2) then pay the short window, the root in the mixing argument and the summable telescoping coefficients. There is no unspecified population-dependent exponent in those choices.

C.3 A numerical exponent ledger

Put $R = R_{\mathrm{num}} \geq 2$. From proposition 3.1, all its numerical outputs are bounded by $2^{1408}R^{128}$. The following successive bounds follow by substitution; they are intentionally coarse.

Quantity	Power-of-two factor	Power of R
M	2^{3}	2
L_*	2^{28}	13
J	2^{1437}	142
a^{-1}	2^{1500}	163
ρ^{-1}	2^{3000}	326
Z_*	2^{3001}	326
M	$2^{3\,074\,048}$	333 824
I	$2^{18\,444\,288}$	2 002 944
W	$2^{196\,739\,072}$	21 364 736
h	$2^{18\,446\,000}$	2 003 200
$H, H\tau, \max_e c_e^{-1}$	$2^{18\,448\,000}$	2 003 400
δ^{-1}	$2^{2\,000\,000\,000}$	200 000 000

For example $8/v = r^{-3}$ and $q^{-1} \leq 2^{12}\kappa^{-6}$ give the R^{13} proposal bound. In the stationary-bias radius term, J contributes degree 142, B_* degree 13, $(1+N)^4$ degree four, and $\epsilon^{-1}\kappa^{-3}$ degree four. Their sum is 163. Squaring gives degree 326 for ρ^{-1}. Applying the fixed powers $1024, 6, 64$ gives the next three rows. The ninth power in (7.3) still has degree below 200 000 000. Thus every row, the activity bounds, and the resource estimates at $H\tau$ are covered by CR^p with the C, p in theorem 7.1.

C.4 Algorithm termination and binary accounting

All emitted species and channels are enumerated directly from the expanded lists, the reverse involution and the structural rejection rule. Their number is polynomial in those lists. The paired kinetic algorithm has the binary bound in proposition 3.1. Following it, the calculations use fixed-dimensional rational geometry, rational arithmetic, integer ceilings, comparisons and fixed integer powers. The powers q_r and $2D+2$ in (2.3) are bounded by the expanded input size. Rational input magnitudes are at most exponential in their binary description, so $\log B$, $\log U$, $\log \Gamma$ and $\log \Xi$ are polynomial in L. Every later fixed power preserves polynomial bit length, with a universal, possibly enormous constant. Ordinary exact arithmetic therefore terminates within a universal polynomial bound; the more conservative mixed bound $C(1+L+R)^p$ covers it and all numerical physical charges simultaneously.

The algorithm never enumerates the chemical class, the fine fibres or the hard-ball configuration space. The analytic proof may use its finite-state kinetic fibres, Sobolev forms and minimizers without making them algorithmic inputs. This distinction is necessary: an existence proof for a stationary distribution or a PDE solution would not itself be a compiler.

D Exact import contracts and dependence

The kinetic theorem. We use Theorem 9.1 of the 13 November 2026 version of [6], not its general open-network Theorem 2.1. Its inputs are a finite labelled expanded table with positive rational rates, the prescribed reverse involution and rational activities, positive integer reaction-wise conservation, a binary initial count, duration and accuracy. Equations (2.1)–(2.2) verify these hypotheses. Invoking it at $\epsilon/8$ makes its cutoff exactly (2.3); the remaining numerical charge is

at most $8R_{\mathrm{num}}$. Its actual output and path guarantee are stated in proposition 3.1. The kinetic theorem's proof uses no assertion of the present paper. Thus the dependency is one way.

The derivative formula. Cheng–Zhang [3, Theorem 2.5, equations (2.9)–(2.10)] applies to a nonexplosive reflecting diffusion $\Delta_{g_t}+\mathcal{Z}_t$ on a smooth manifold with smooth time-dependent complete metrics. It assumes lower bounds K for $\mathrm{Ric}-\nabla\mathcal{Z}-\frac{1}{2}\partial_t g$ and σ for the boundary second fundamental form, and finiteness of the exponential of the negative curvature and boundary-local-time integrals. For smooth terminal data it also assumes the intervening gradients bounded. It provides the derivative representation with any C^1 control h having $h(s)=0$, $h(t)=1$, and a multiplicative functional bounded by the same exponential. We use its linear control. In our application smooth compact approximants are nonexplosive and complete as manifolds with boundary; (A.14) identifies the generator, (A.15) and the clock moments verify the integrability, and smooth Neumann data satisfy the bounded-gradient hypothesis. Uniform estimates permit the two smoothing limits described above. The theorem is not applied directly to a nonsmooth polytope or a merely Lipschitz time profile.

Contextual results, rather than extra hidden hypotheses. Fradon [4, Theorems 2.2, 2.4 and Proposition 3.4] proves normal-reflection globule results with bounded Lipschitz coefficients and compatible constraint geometry, and discusses reversible gradient systems. It does not by itself give our time-dependent variable-conormal law. Our core-potential and approximation argument supplies that passage. Zhang–Isaacson [7, Section III, equation (III.2), Theorem III.1] express spatial balance as an equality of forward/reverse reaction fluxes. Proposition 3.2 verifies that equality directly for our ordered slots and placement kernels. Hoessly–Wiuf's finite-closed-set reduction [5, Assumption 2, Theorem 4.1] and the Poisson law theorem of Anderson–Craciun–Kurtz [1, Theorem 4.1] are comparisons; neither is substituted for our marked-path coupling or for a hard-core count normalizer. The experimental work [2] supplies no mathematical rate, leakage or material-realization premise here.

References

[1] D. F. Anderson, G. Craciun and T. G. Kurtz, *Product-form stationary distributions for deficiency zero chemical reaction networks*, arXiv:0803.3042v4 (2010). https://arxiv.org/abs/0803.3042v4.

[2] Y.-J. Chen et al., *Programmable chemical controllers made from DNA* (2013). https://pmc.ncbi.nlm.nih.gov/articles/PMC4150546/.

[3] L.-J. Cheng and K. Zhang, *Reflecting diffusion semigroup on manifolds carrying geometric flow*, arXiv:1211.3623v2 (2017). https://arxiv.org/abs/1211.3623v2.

[4] M. Fradon, *Brownian dynamics of globules*, arXiv:0910.5394v1 (2009). https://arxiv.org/abs/0910.5394v1.

[5] L. Hoessly and C. Wiuf, *Fast reactions with non-interacting species in stochastic reaction networks*, arXiv:2110.13235v1 (2021). https://arxiv.org/abs/2110.13235v1.

[6] H. Sandhu, *Compiling stochastic chemical trajectories with elementary reactions*, version 1.0, 13 November 2026, Theorem 9.1. Zenodo, doi:10.5281/zenodo.24170339.

[7] Y. Zhang and S. A. Isaacson, *Detailed balance for particle models of reversible reactions in bounded domains*, arXiv:2201.04099v3 (2022). https://arxiv.org/abs/2201.04099v3.